MATH MA’AM ACADEMY

Where Every Subject Clicks.

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UNIT 10: STATIC ELECTRICITY

Study Notes | Based on the textbook pages provided

1. Static Electricity

• **Static electricity** is electric charge at rest on the surface of a body.

• Rubbing two suitable materials can transfer electrons from one material to another, leaving the bodies electrically charged.

• Nylon or synthetic clothes may cling to the body and produce tiny sparks or crackling sounds when removed quickly. This is due to static charge.

• Static charge can develop on conductors as well as insulators.

2. Current Electricity

• **Current electricity** is the flow of electric charge through a conductor in a complete circuit.

• In metal wires, the moving charge carriers are electrons. Electron flow is from the negative terminal towards the positive terminal.

• **Conventional current** is defined to flow from the positive terminal to the negative terminal in the external circuit.

• Current electricity is used in lighting, heating, refrigeration and operating electrical appliances.

3. Electric Cells and Batteries

• An **electric cell** converts chemical energy into electrical energy and acts as a source of electricity.

• **Dry cell:** A common primary cell. In the textbook description, the zinc container is the negative terminal and the metal cap is the positive terminal.

• **Primary cell:** Its chemical reaction is not readily reversible; when the chemicals are used up, the cell must be replaced.

• **Secondary cell:** It can be recharged because its chemical reaction is reversible under suitable conditions.

• **Button cell:** A small cell used in devices such as watches, calculators and hearing aids.

• **Battery:** Two or more cells connected together. The positive terminal of one cell is connected to the negative terminal of the next in a series arrangement.

• **Solar cell:** Converts light energy directly into electrical energy; it does not use a chemical reaction as its energy source.

4. Static Charge and Charging by Rubbing

• When a plastic ruler is rubbed with a woollen sleeve, it may attract small pieces of paper because it becomes electrically charged.

• Rubbing does not create electric charge from nothing; it transfers electrons between materials.

• The two bodies acquire equal amounts of charge with opposite signs in the idealised charging-by-friction model.

• A glass rod rubbed with silk becomes positively charged; an ebonite rod rubbed with wool becomes negatively charged.

5. Nature of Electric Charges

• There are two kinds of electric charge: **positive** and **negative**.

• **Like charges repel** one another: positive–positive and negative–negative.

• **Unlike charges attract** one another: positive–negative.

• A body becomes positively charged when it loses electrons; it becomes negatively charged when it gains electrons.

6. Electroscope

• An **electroscope** is an instrument used to detect small electric charges on a body.

• A gold-leaf electroscope is sensitive and can detect even a small amount of charge.

• Uses: (1) to test whether a body is charged, (2) to identify the kind of charge when compared with a known charge, and (3) to investigate whether a material is a conductor or an insulator.

7. Conductors and Insulators

• **Conductors** allow electric charge to pass through them relatively easily. Examples: copper, iron, gold, silver and aluminium.

• **Insulators** do not allow electric charge to pass through them easily. Examples: glass, leather, ebonite, paper and many plastics.

• Metals are generally good conductors because they contain mobile electrons.

8. Static Electricity vs Current Electricity

Static electricity	Current electricity
Charge accumulated at rest on a body.	Charge flowing through a conducting path.
Can occur on conductors and insulators.	Sustained current requires a conducting path and a potential difference.
Detected using an electroscope.	Measured using an ammeter; voltage is measured using a voltmeter.
Examples: crackling clothes and lightning.	Examples: current in a bulb, fan or other electrical appliance.

9. Important Exam Points

• A cell converts chemical energy into electrical energy; a solar cell converts light energy into electrical energy.

• A primary cell is generally non-rechargeable; a secondary cell is rechargeable.

• A battery is made by connecting two or more cells.

• Like charges repel; unlike charges attract.

• A glass rod rubbed with silk becomes positive; an ebonite rod rubbed with wool becomes negative.

• Do not confuse electron flow with conventional current: they are in opposite directions in a metal wire.

ALL THE BEST